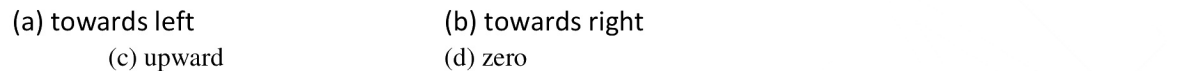


(a) towards left
(b) towards right
(c) upward
(d) zero



(a) towards left
(b) towards right
(c) upward
(d) zero



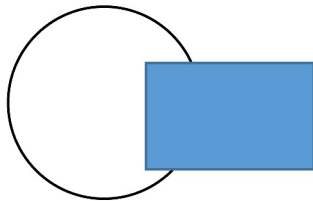
7. Figure shows two conducting thin concentric shells of radii r and $3r$. The outer shell carries charge q , inner shell is neutral. The amount of charge which flows from inner shell to the Earth after the key K is closed is equal to
- 
- The diagram shows two concentric circles representing the shells. The inner circle has a radius labeled r and the outer circle has a radius labeled $3r$. The outer circle is labeled with a charge q . A key, labeled K , is connected to the inner shell. The key is shown in an open position, with one terminal connected to the inner shell and the other terminal connected to a ground symbol (three horizontal lines of decreasing width).



- (a) $\phi_1 = 0$ (b) $0 < \phi_1 < \phi$
(c) $\phi_1 = \phi$ (d) $\phi_1 > \phi$

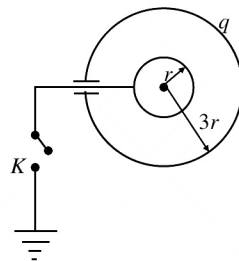
A thin, metallic spherical shell contains a charge Q on it. A point charge q is placed at the center of the shell and another charge q_1 is placed outside it as shown in figure. All the three charges are positive. The force on the charge at the center is

$$(x + a)^n = \sum_{k=0}^n \binom{n}{k} x^k a^{n-k}$$



- (a) towards left
(b) towards right
(c) upward
(d) zero

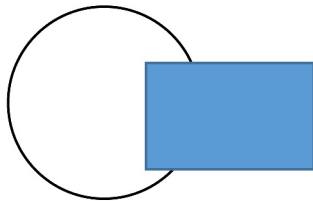
7. Figure shows two conducting thin concentric shells of radii r and $3r$. The outer shell carries charge q , inner shell is neutral. The amount of charge which flows from inner shell to the Earth after the key K is closed is equal to



- (a) $-\frac{q}{3}$
(b) $\frac{q}{3}$
(c) $3q$
(d) $-3q$
- (a) $\phi_1 = 0$
(b) $0 < \phi_1 < \phi$
(c) $\phi_1 = \phi$
(d) $\phi_1 > \phi$

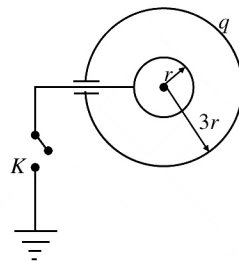
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- (a) $\phi_1 = 0$
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